

Gap 1: Static Utility + Group-Blind Fairness

The paper's utility formula only uses distance: $Utility = Geo(dropoff, pickup) - Geo(pickup, driver_location)$. It explicitly states this stays the *same regardless of time* — so a 5km trip at 2 PM and a 5km trip at peak 6 PM traffic are valued identically. Real traffic/time cost is ignored.

Also, their fairness metric $Var(driver\ earnings)$ treats all drivers equally regardless of hours worked.

Example: If Driver A drives 60 hrs/week and earns ₹6000, and Driver B (part-timer) drives only 15 hrs/week and also earns ₹6000 — the paper calls this "perfectly fair" (variance = 0). But Driver A's real rate is ₹100/hr while Driver B's is ₹400/hr. That's a hidden 4x gap the paper's metric can't see.

Proposed fix: Make utility time/traffic-aware (reuse their existing demand-prediction module for this), and split drivers into activity groups (full-time/part-time), then compare fairness within and between these groups **using hourly rate**, not raw totals.

Gap 2: Utilization/Access Fairness

The paper only checks final earnings, never checks if an available driver was actually given work.

Example: Driver X is online 8 hrs but assigned only 2 trips (idle 75% of time). Driver Y is online 8 hrs and assigned 10 trips. If X's 2 trips happen to be high-value, both can end up with equal total earnings — paper says "fair." But X wasted 6 hours waiting with no assignment — that's a real inequality the paper never measures.

Proposed fix: New metric — $Utilization = \text{assigned time-slots} / \text{online time-slots per driver}$, and check its variance across drivers/groups.